

Detailed Project Summary

Project Name -Enterprise HR Analytics and Attrition Risk Analysis

Project Overview - This project simulates an enterprise HR analytics workflow designed to identify workforce attrition drivers using business-driven synthetic employee data.

Dataset- 15,000 employee records- 30+ HR attributes- Synthetic enterprise-style dataset with realistic business relationships.

Project Workflow

1. Business Logic Design

- Avoided using public datasets from platforms like [Kaggle](#) since they are small, clean, and less representative of real business data.
- Studied real-world HR attrition reports and common workplace patterns to identify major employee attrition drivers.
- Selected key factors affecting attrition such as Salary, Overtime, Job Satisfaction, Manager Rating, Performance Rating, and Work Conditions.
- Initially generated fully random data, but analysis produced no meaningful insights, which halted project progress.
- Redesigned data architecture by introducing realistic business relationships between important variables.
- Created dependencies such as:
 - Higher Overtime → Lower Job Satisfaction
 - Lower Salary → Higher Attrition Probability
 - Poor Manager Rating → Lower Satisfaction
 - Lower Performance Rating → Higher Attrition Risk
- Built Attrition variable using a risk scoring model instead of random assignment, where multiple factors contributed to employee exit probability.

2. Synthetic Data Generation

- Generated a synthetic enterprise-level HR dataset using Python, Pandas, and NumPy.
- Created 15,000 employee records to simulate large organizational workforce data.
- Designed dataset with 30+ HR attributes including EmployeeID, Department, Salary, Overtime, Satisfaction, Performance, Manager Rating, Attrition, etc.

- Structured data generation to preserve business relationships defined during design stage.
- Used synthetic data generation instead of static datasets to simulate realistic enterprise analytics environment.

3. Data Quality Simulation

- Intentionally introduced real-world data quality issues to simulate enterprise data challenges.
- Added duplicate records to test duplicate detection process.
- Introduced missing/null values across multiple columns.
- Created inconsistent categorical values to simulate real database inconsistencies.
- Purpose was to replicate messy organizational data before analysis stage.

4. Data Cleaning & Preprocessing

- Removed duplicate employee records to maintain data integrity.
- Standardized inconsistent department/category labels.
- Faced challenge in selecting correct imputation methods for missing values without distorting business relationships.
- Applied statistical imputation techniques instead of random replacement:
 - Department-wise Median Salary Imputation
 - Department-wise Manager Rating Imputation
 - Median-based Job Satisfaction Imputation
- Ensured original business logic remained preserved after cleaning.

5. Data Dictionary Creation

- Created complete data dictionary for all variables in the dataset.
- Documented column definitions, data types, and business meaning of every attribute.
- Improved dataset readability and maintained structured project documentation.
- Simulated documentation standards followed in real analytics projects.

6. Exploratory Data Analysis (EDA)

- Performed business-driven EDA instead of random statistical exploration.
- Major challenge was analyzing a dataset with more than 30 variables and deciding analytical direction.
- Broke larger problem of employee attrition into smaller business segments for structured analysis.
- Created analysis flow across multiple segments such as:
 - Compensation Analysis
 - Employee Satisfaction Analysis
 - Work Environment Analysis
 - Performance Analysis
 - Managerial Impact Analysis
- Tested business hypotheses under each segment to identify meaningful workforce behavior patterns.

7. Business Insights Generation

- Identified strongest employee retention and attrition drivers through segmented analysis.
- Analyzed impact of compensation, overtime workload, managerial quality, satisfaction, performance, and work conditions.
- Focused on generating business-level workforce insights rather than isolated statistical observations.
- Derived patterns explaining why employees leave organizations and what factors influence retention.

8. Dashboard Design & Storytelling

- Built interactive dashboard to present workforce analytics insights visually.
- Major challenge was selecting the correct chart type for each insight.
- Focused on maintaining continuous business storytelling instead of showing disconnected charts.

- Designed dashboard flow from high-level attrition overview to deeper root cause analysis.
- Structured visualizations to support management-level decision making.

Project Outcome

- *Simulated an end-to-end real-world HR analytics project pipeline.*
- *Worked on synthetic data generation, data engineering, cleaning, EDA, business insights, and dashboard development.*
- *Gained practical experience in handling realistic enterprise data challenges beyond traditional public dataset projects.*
- *Successfully created a business-focused analytics workflow replicating real organizational decision-making process.*

Key Technologies- Python- Pandas- NumPy- Power BI

Major Insights- Overtime and burnout are strongest attrition drivers- Lower job satisfaction strongly predicts attrition- Engagement score strongly predicts retention- Compensation and bonus structures influence retention- Poor managerial effectiveness increases attrition risk- Promotion opportunities improve retention